EXPERIMENT NO. 9

Experiment Name: Implementation of Subnetting in Cisco Packet Tracer Simulator

Aim

To design and implement multiple subnets using Cisco Packet Tracer, assign static IP addresses to hosts and router interfaces, and establish communication between different subnets through a router. To simulate packet transmission with and without a router and observe the importance of routing in interconnecting different networks.

Procedure

1. Open Cisco Packet Tracer and create a new network topology. Select the required networking devices from the device panel, including four PCs, one switch for each subnet, and a router to connect the two networks.
2. Place PC0 and PC1 on the left side of the workspace and connect them to Switch0 using copper straight-through cables. Place PC2 and PC3 on the right side and connect them to Switch1 in the same manner.
3. Connect Switch0 to the first router interface and Switch1 to the second router interface using copper straight-through cables. Wait until the connection indicators turn green, showing that the interfaces are active.
4. **Configure the first subnet:** Click PC0, select Desktop, and open IP Configuration. Assign the following details:
    - IP address: 192.168.1.2
    - Subnet mask: 255.255.255.0
    - Default gateway: 192.168.1.1
5. Configure PC1 with IP address 192.168.1.3, subnet mask 255.255.255.0, and default gateway 192.168.1.1. These two PCs belong to the first subnet, 192.168.1.0/24.
6. **Configure the second subnet:** Click PC2 and enter IP address 192.168.2.2, subnet mask 255.255.255.0, and default gateway 192.168.2.1. Configure PC3 with IP address 192.168.2.3, subnet mask 255.255.255.0, and default gateway 192.168.2.1. Both PCs belong to the second subnet, 192.168.2.0/24.

7. **Configure the router:** Click the router, select the CLI tab, and enter the following commands to configure the interface connected to the first subnet:

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

8. Configure the second router interface for the second subnet:

interface gigabitEthernet 0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

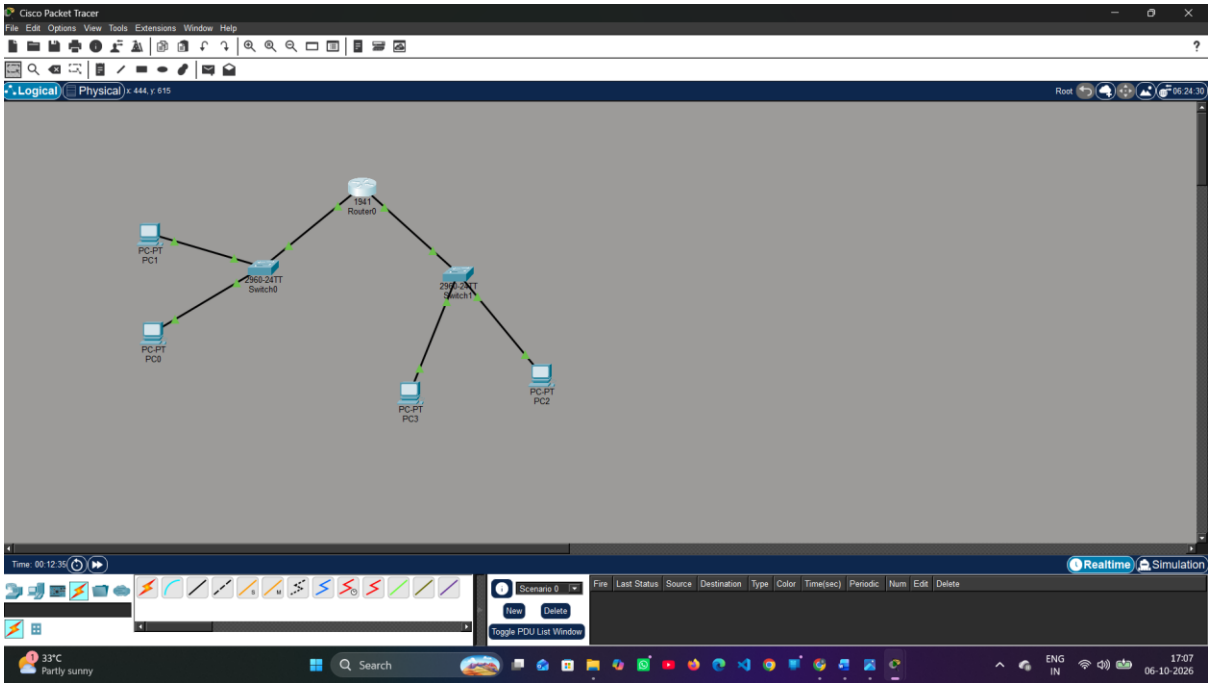
exit

end

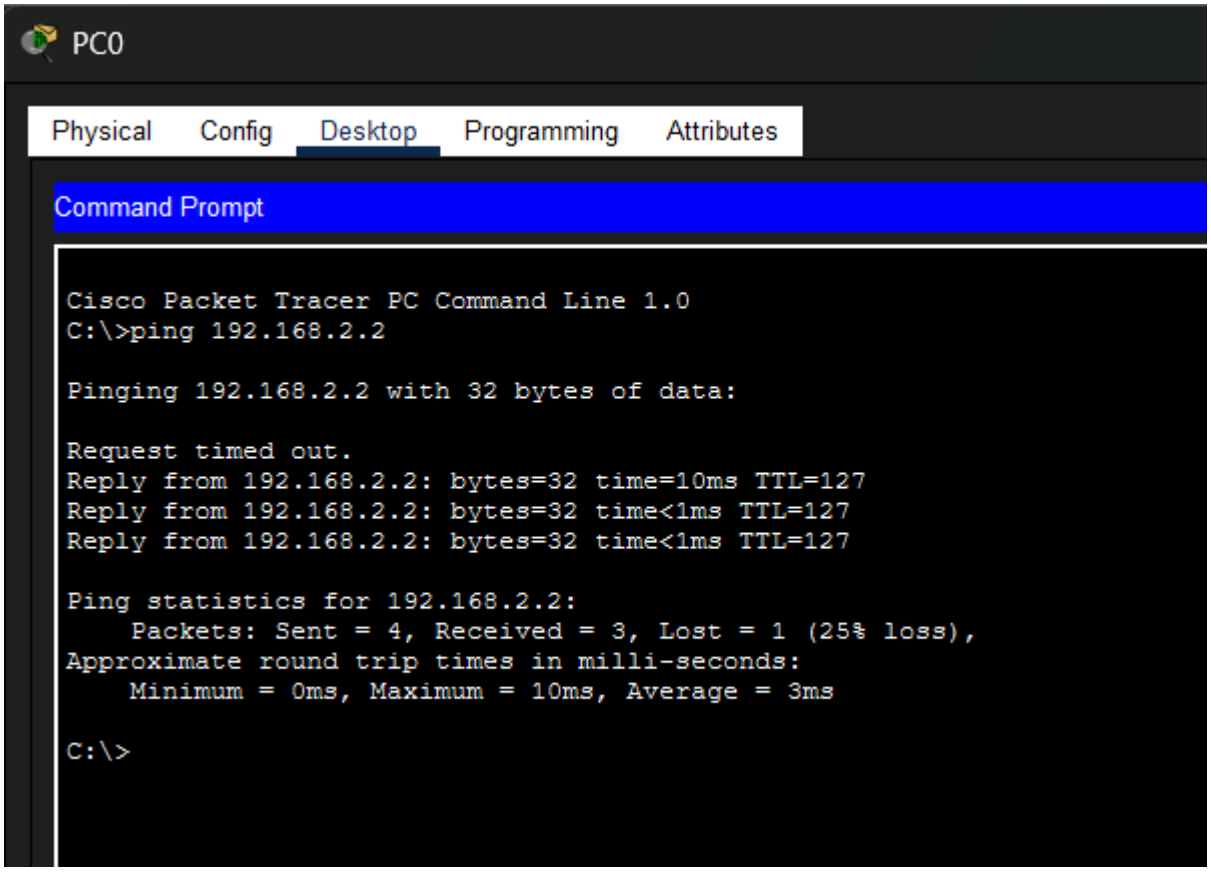
The router now acts as the gateway for both subnets and forwards packets between them.

9. **Test communication within a subnet:** Click PC0, select Desktop, and open Command Prompt. Execute ping 192.168.1.3 to test communication between PC0 and PC1. Repeat the test from PC2 to PC3 using ping 192.168.2.3.
10. **Test communication between subnets:** From PC0, execute ping 192.168.2.2 and ping 192.168.2.3. Observe the responses. If the IP addresses, gateways, router interfaces and connections are configured correctly, packets should reach the destination PCs through the router. The first ping may time out while address resolution takes place; repeat it if necessary.
11. **Observe communication without a router:** Disconnect the router from the topology or disable the relevant router interfaces. Attempt to ping a PC in the other subnet again. Observe that communication between the two subnets fails because there is no active routing path between them. Communication within each individual subnet may still work.
12. Compare the ping results obtained with the router enabled and disabled. Observe the packet transmission in Simulation mode by selecting the Simulation tab, generating a ping, and using Capture/Forward to examine how packets move between the PCs, switches and router.

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Result

Thus, multiple subnets were designed and configured successfully in Cisco Packet Tracer using static IP addressing. Communication within each subnet and between different subnets was tested using the ping command. The simulation demonstrated that a router is required to forward packets between different IP networks and that inter-subnet communication fails when the routing path is unavailable.